



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

ASSIGNMENT 1

SECP2753

SECTION 02

DATA MINING

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TABLE OF CONTENT

CONTENT	PAGE
1. Introduction of dataset	3
2. Dataset characteristics	4
3. Insights	
3.1 Insight number of students per grade	5
3.2 Insight number of female and male students per grade	6
3.3 Insight number of students per department	7
3.4 Insight number of students per grade per department	8
3.5 Insight of family income level per grade	9
3.6 Insight of average total score vs stress level	10
4. Conclusion	11
5. Reference	12

Introduction

According to Larose D., the Cross-Industry Standard Process for Data Mining (CRISP-DM) is a standard process used that covers entire process, from communication of the business problem until the model deployment. The business insights are one of the steps focusing obtaining the objectives and requirements from a business perspective.

The dataset used for this report consists of 1,000 student records collected from four academic departments that are Mathematics, Computer Science, Business, and Engineering. The attributes included in the dataset are varied. For instance, gender, age, grade, department, sleep duration, stress level, family income, and parental education.

This report aims to uncover meaningful insights and information including patterns related to the dataset. This is to understand in-depth the student performance and behavioral factors. By using visualizations such as bar charts and pie charts, we can analyze how grades are distributed, how gender affects academic outcomes, how departmental representation varies, and how factors like sleep and stress impact academic performance

Dataset Characteristics

The dataset contains 5,000 student records with the following characteristics:

1. Personal Information: Student_ID, First_Name, Last_Name, Email, Gender, Age, Department
2. Academic Performance:
 - a. Scores: Midterm_Score, Final_Score, Assignments_Avg, Quizzes_Avg, Participation_Score, Projects_Score
 - b. Aggregate: Total_Score, Grade
3. Attendance and Study Habits: Attendance (%), Study_Hours_per_Week, Sleep_Hours_per_Night
4. Background Factors: Extracurricular_Activities, Internet_Access_at_Home, Parent_Education_Level, Family_Income_Level
5. Wellbeing: Stress_Level (1-10)

3.1 Insight number of students per grade

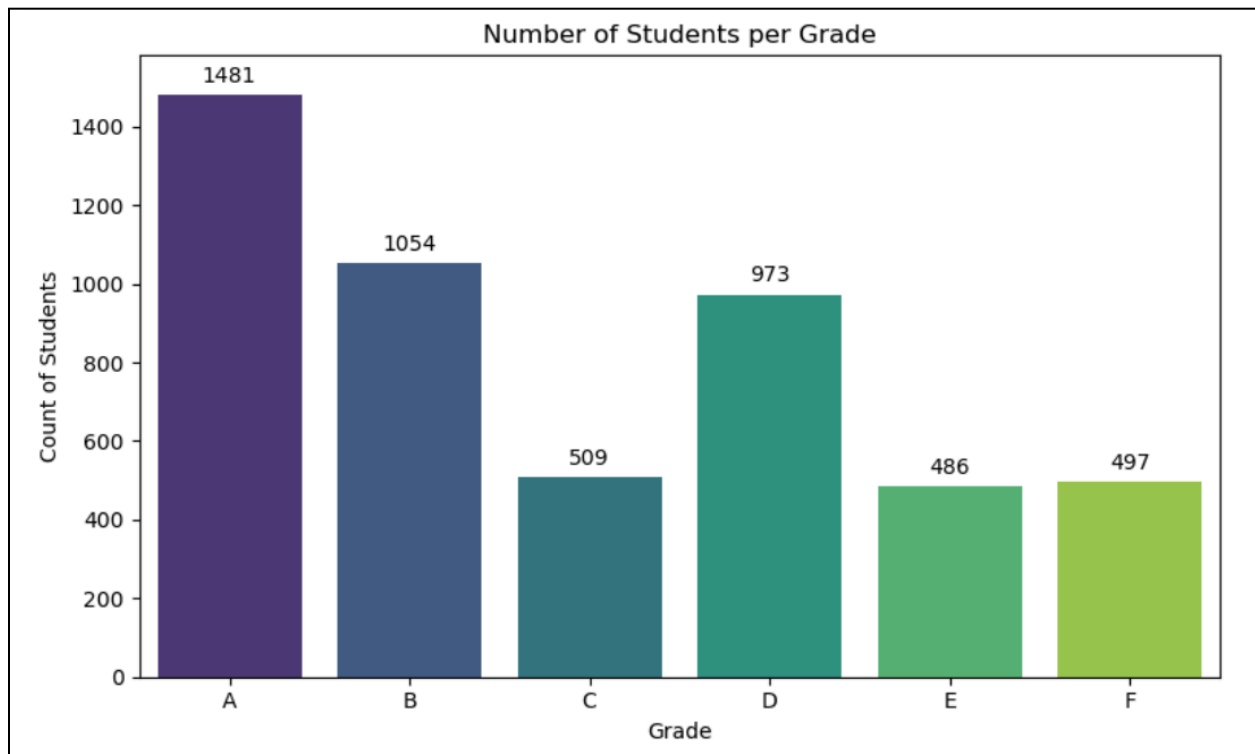


Figure 3.1

Based on Figure 3.1, it shows a bar plot of number of students per grade. The attribute that we used for this insight is Grade. Based on the data that we have cleaned and analyzed, it shows that Grade A has 1481 students, Grade B has 1054 students, Grade C has 509 students, Grade D has 973 students, Grade E has 486 students, and grade F has 497 students. The key insight of this finding is it shows that there are only 29.62% students who managed to get an A.

3.2 Insight number of female and male students per grade

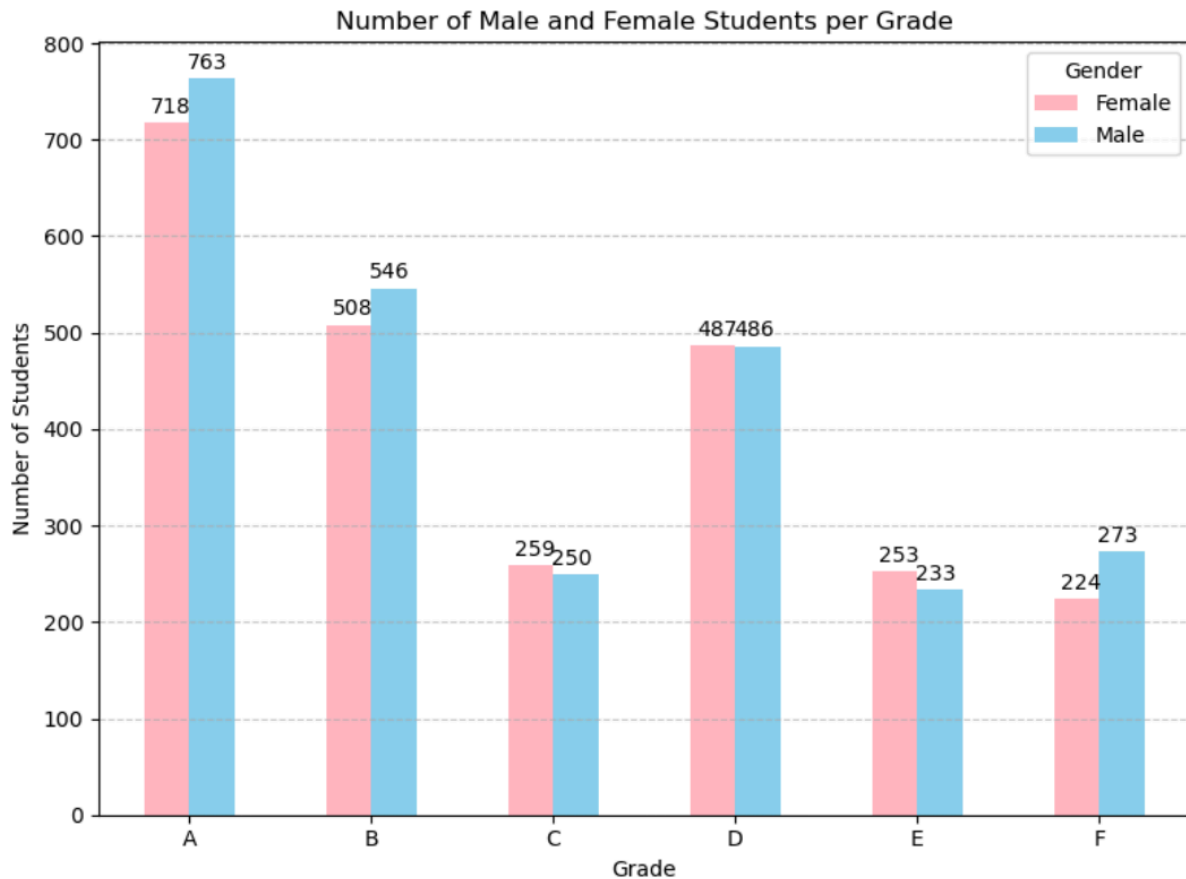


Figure 3.2

Based on Figure 3.2, it shows a bar plot of the number of male and female students for each grade. The Female is indicated by a box in pink colour, meanwhile the male is indicated by a box in blue colour. The attributes that we used for this insight is grade and gender. For grade A, male students slightly outperform females by 45 students in this grade. For grade B, male students outnumber females students again. Then, for grade C, shows that female students slightly outnumber male students. Next, for grade D, it shows that male and female are almost equal numbers, suggesting a consistent distribution between genders in average performers. For grade E, females exceed males, and lastly for grade F, male students exceed females, indicating that more males are failing.

The key insight of this finding is that Males dominate at both extremes. They have higher counts in the top grade (A) and the lowest grade (F). Then, female students show more clustering in the mid-to-lower ranges, especially in grades C and E. and lastly Grade D is almost gender-neutral in terms of student count.

3.3 Insight number of students per department

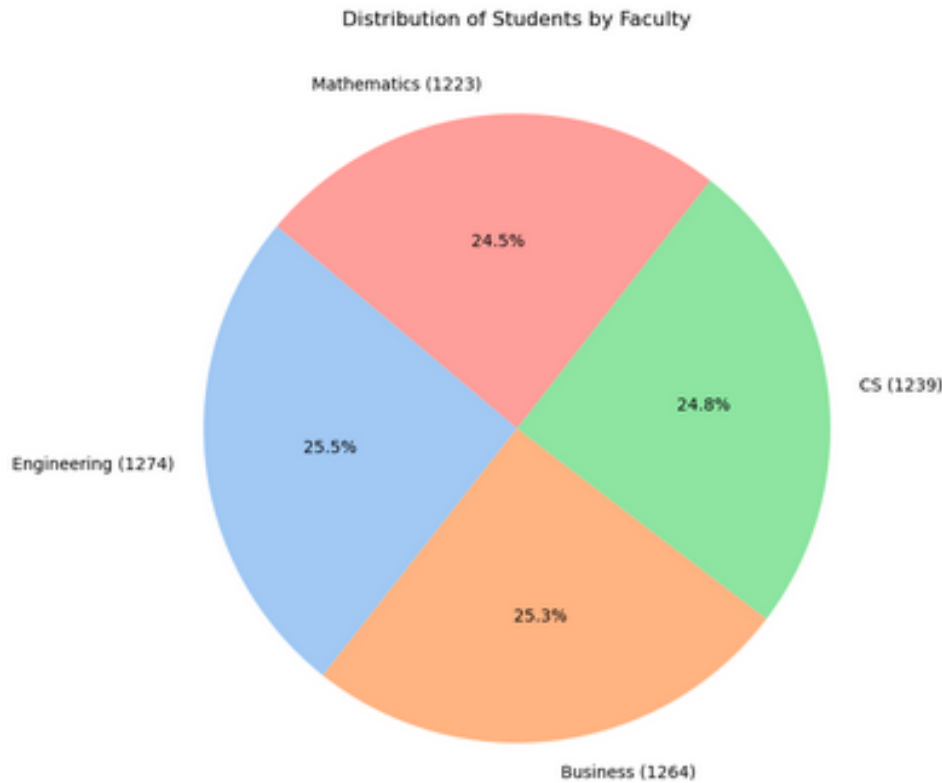


Figure 3.3

Figure 3.3 shows a pie chart for the division of the number of students by department. There are four departments included in the dataset which are Mathematics, Computer Science, Business and Engineering. The department of Mathematics consists of 1223 students, Computer Science department has 1239 students, Business department has 1264 students and the department of Engineering consists of 1274 students.

Based on the pie chart, the Engineering department has the most number of students, making up 25.5% of the total, followed by the Business department which is 25.3% of the total. Next, the Computer Science department which is 24.8% of the total, and the lowest number of students is the department of Mathematics which is 24.5% of the total. This shows that the differences in the number of students according to the department are relatively small and the dataset is balanced.

3.4 Insight number of students per grade per department

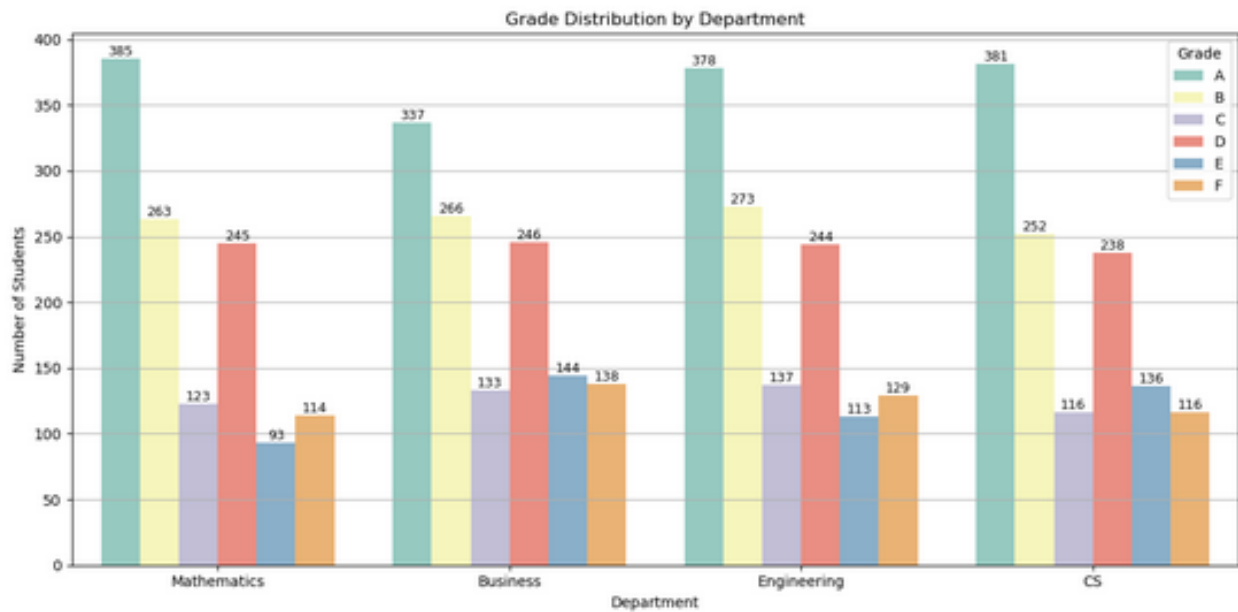


Figure 3.4

Figure 3.4 presents the number of student grades across four departments which are Mathematics, Business, Engineering, and Computer Science. Grade A is the most frequently achieved grade in all departments, with Mathematics having the highest number of 385 students, followed closely by Computer Science with 381 students, Engineering with 378 students, and Business with 337 students. Grade B is also commonly achieved by students. Department of Engineering has the highest number of B grades which are 273 students, while Computer Science shows the lowest number of B grades which are 252 students.

Grade E and Grade F reflect poorer performance but they appear less frequently across the four departments. The Business department has a slightly higher count of E and F grades consisting of 144 and 138 students each. Meanwhile, the Mathematics department has the lowest number of E and F grades consisting of 93 and 114 students each.

3.5 Insight of average total score by sleep hours per night

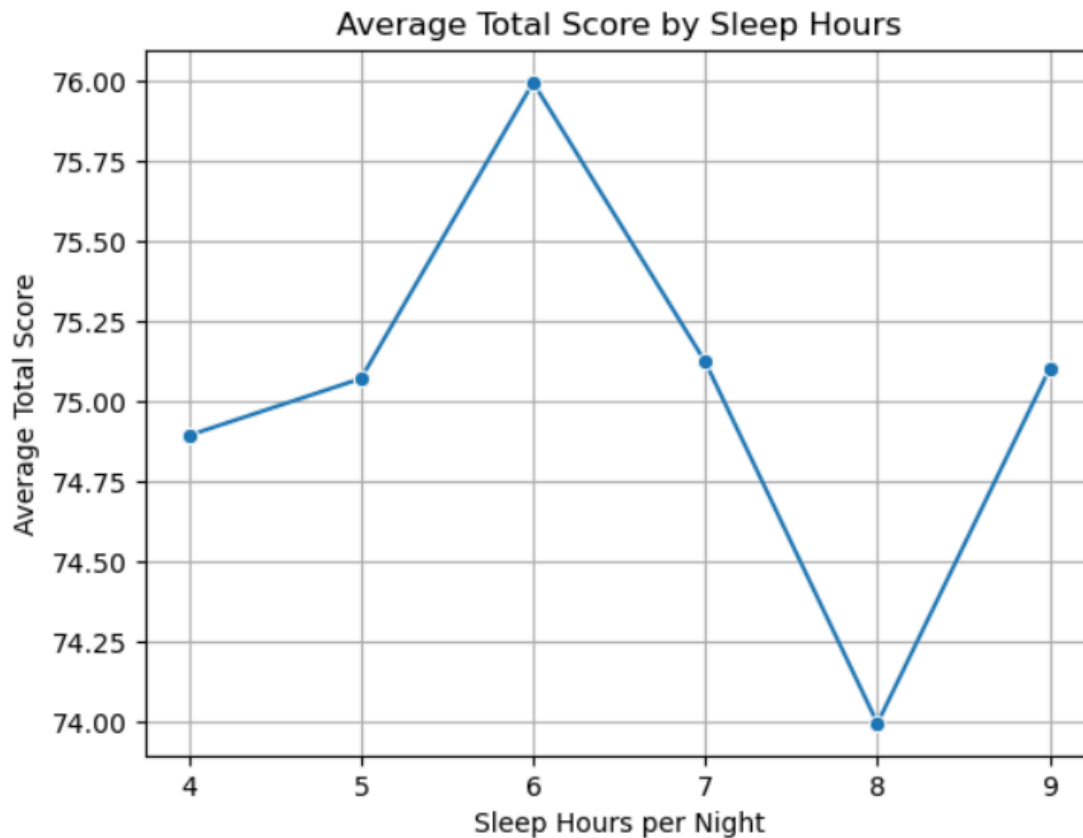


Figure 3.5

Figure 3.5 shows the non-linear relationship between students' sleep duration and academic performance. The highest average scores are observed among students who sleep for 6 hours per night on average which suggests this may be the optimal amount of rest for peak cognitive function. Interestingly, performance declines beyond this point, reaching the lowest average at 8 hours of sleep. This challenges the assumption that more sleep always leads to better outcomes.

In fact, students who sleep excessively may experience diminished academic performance, possibly due to underlying issues like fatigue, lack of motivation or poor time management. On the other end, students that sleep for around 4-5 hours show moderately low scores, likely due to sleep deprivation affecting focus and memory. These findings suggest that both insufficient and excessive sleep may negatively impact academic outcomes, while a balanced amount of rest may offer the most academic benefits.

3.6 Insight of average total score vs stress level

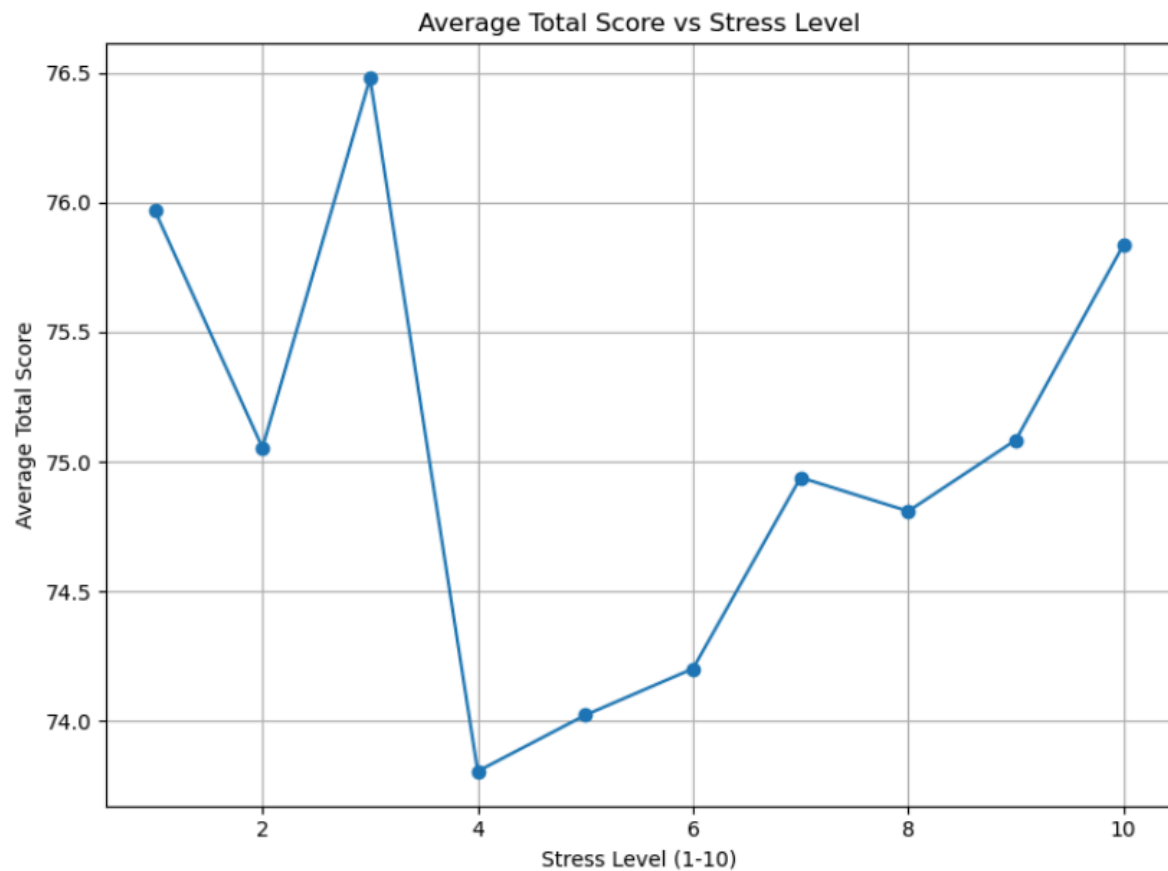


Figure 3.6

Figure 3.6 illustrates a non-linear relationship between stress levels and academic performance. Students experiencing either low or high levels of stress tend to achieve higher average scores, while those with moderate stress levels (ranging from 4 to 6) consistently perform lower. This suggests that a certain degree of stress may serve as a motivational force, pushing students to stay focused and perform well. In contrast, stress that is poorly managed may hinder performance, rather than help it.

Notably, students within the moderate stress range appear to be in a less advantageous position. These individuals may feel the weight of academic pressure, yet lack the coping mechanisms or drive needed to channel that stress into productive effort. As a result, their stress may manifest as procrastination, reduced concentration, or self-doubt, all of which can detract from academic success.

Conclusion

In this assignment, we explored various dimensions of students' data and performance using data mining techniques to unleash meaningful patterns and insights. By analyzing the dataset, we observed trends related to grades, gender distribution, departmental representation and other influential factors such as stress levels, sleep duration, family income and parental education.

Our findings revealed that academic performance is shaped by a complex interplay of factors. Notably, stress and sleep displayed non-linear relationships with academic scores moderate stress and excessive sleep appeared to negatively impact performance, whereas optimal stress and rest levels were associated with higher achievement. Additionally, gender differences were apparent, particularly in the highest and lowest grades, and departmental distribution showed a relatively balanced dataset, indicating fair representation.

These insights highlight the importance of considering both academic and non-academic factors when assessing student outcomes. Understanding how lifestyle, psychological, and demographic factors affect performance can help educators and policymakers create more supportive learning environments. Ultimately, this report emphasizes that data mining is a powerful tool for identifying trends and guiding data-driven decisions in education.

REFERENCE

1 <https://www.kaggle.com/datasets/mahmoudelhemaly/students-grading-dataset>

2 Larose, D. T., & Larose, C. D. (2014). *Discovering knowledge in data : an introduction to data mining*. Wiley.